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# Niyonkuru Ally Sun Boduen

FULL-STACK DEVELOPER

AI ENGINEER

ML ENGINEER

IOT & EMBEDDED SYSTEMS

☎0792652471 ☎0732972867 ✉sunboduen@gmail.com

4+

YEARS EXP.

15+

PROJECTS

13+

LANGUAGES

5

DOMAINS

## SKILLS

### WEB DEVELOPMENT

|                 |      |
|-----------------|------|
| HTML / CSS      | 100% |
| JavaScript / TS | 100% |
| PHP             | 100% |
| Python          | 100% |
| Node.js         | 100% |

### FRAMEWORKS

|                      |      |
|----------------------|------|
| React.js / Next.js   | 100% |
| Vue.js               | 100% |
| Django               | 100% |
| Laravel              | 100% |
| Tailwind / Bootstrap | 100% |

### AI / MACHINE LEARNING

|                  |      |
|------------------|------|
| TensorFlow       | 100% |
| Machine Learning | 90%  |
| Deep Learning    | 85%  |
| Neural Networks  | 82%  |

### DATABASES

|         |      |
|---------|------|
| MySQL   | 100% |
| MongoDB | 100% |
| Neo4j   | 100% |

### IOT & EMBEDDED

|              |      |
|--------------|------|
| Arduino      | 100% |
| Raspberry Pi | 100% |
| Robotics     | 100% |
| Blockchain   | 80%  |

## LANGUAGES

|             |       |
|-------------|-------|
| Kinyarwanda | ■■■■■ |
| English     | ■■■■■ |
| French      | ■■■■■ |

CONTACT

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- 0732972867 (WhatsApp)
- sunboduen@gmail.com

Open to exciting full-time, contract, and freelance opportunities in software engineering & AI.

✦ AVAILABLE FOR HIRE

PROFESSIONAL SUMMARY

Innovative Full-Stack Developer & AI Engineer with over 4 years of hands-on experience architecting and shipping production-grade software across web, mobile, and embedded platforms. Proven expertise in designing scalable, AI-powered applications using modern frameworks and neural network technologies, from concept to deployment. Passionate about bridging the physical and digital worlds through IoT and embedded systems, having built autonomous robotics and smart infrastructure solutions. Combines a sharp engineering mindset with a designer's eye — delivering systems that are not only technically robust but elegant in execution. Driven by a commitment to innovation, code quality, and impactful problem-solving in every project.

WORK EXPERIENCE

Senior Full-Stack Developer

2023 - Present

Tech Solutions Ltd — Kigali, Rwanda

- Led architecture and delivery of 5+ enterprise web applications serving thousands of concurrent users using React.js, Next.js, and Node.js microservice backends.
- Reduced application load time by 42% through advanced code-splitting, server-side rendering optimization, and CDN strategy implementation.
- Designed and deployed RESTful and GraphQL APIs integrated with MongoDB and MySQL databases, maintaining 99.9% uptime SLA.
- Mentored 3 junior developers, establishing team code review standards and CI/CD pipelines via GitHub Actions.

React.js Next.js Node.js TypeScript MongoDB Docker

AI & Machine Learning Engineer

2022 - 2023

Innovate AI Labs — Remote

- Designed and trained deep learning models using TensorFlow and PyTorch for computer vision tasks including object detection and real-time image classification.
- Built an AI content generation pipeline reducing manual content production time by 70% for a digital media client.
- Implemented reinforcement learning algorithms for autonomous vehicle path optimization in simulation environments.
- Deployed ML models to production via FastAPI endpoints, integrated with React frontends for real-time inference.

TensorFlow Python PyTorch FastAPI OpenCV AWS

Embedded Systems & IoT Developer

2021 - 2022

SmartSys Engineering — Kigali, Rwanda

- Engineered firmware for Arduino and Raspberry Pi platforms powering smart building automation systems across 4 commercial installations.
- Developed an end-to-end smart traffic management prototype integrating sensor arrays, real-time data processing, and cloud dashboards.
- Built MQTT-based IoT data pipelines streaming live sensor telemetry to Node.js backends with real-time web dashboards via WebSocket.

Arduino Raspberry Pi C/C++ MQTT Python Node.js

Web & Mobile Developer

2020 - 2021

Freelance / Independent — Kigali, Rwanda

- Delivered 10+ custom web and mobile applications for local businesses using Laravel, Vue.js, and React Native, from wireframe to launch.
- Built e-commerce platforms with payment gateway integration, inventory management, and admin dashboards.
- Designed responsive, pixel-perfect UI/UX using Figma mockups translated into production Tailwind CSS components.

LaravelVue.jsReact NativeMySQLTailwindFigma

FEATURED PROJECTS



Self-Driving Car System

Autonomous vehicle navigation using computer vision, LIDAR sensor fusion, and deep reinforcement learning on Raspberry Pi + TensorFlow.



Automatic Parking System

IoT-powered smart parking with ultrasonic sensors, real-time slot detection, and a React dashboard for facility management.



Smart Traffic Light System

Adaptive traffic control using computer vision to count vehicles and dynamically adjust signal timing, reducing average wait time by 35%.



AI Creator Studio

Full-stack AI content platform with text, image, and code generation capabilities, custom fine-tuning pipeline, and user workspace management.



Mobile App Suite

Cross-platform React Native applications for fintech and e-commerce clients with offline support, push notifications, and payment integration.



IoT Smart Home Automation

Raspberry Pi-based home automation hub controlling lighting, climate, and security cameras via voice commands and a custom web app interface.

EDUCATION



B.Sc. Computer Science & Engineering

University of Rwanda — College of Science & Technology  
2019 - 2023 · Kigali, Rwanda



Advanced Certificate in AI & Deep Learning

Coursera / DeepLearning.AI Specialization  
2022 · Online

CERTIFICATIONS

|                                       |                     |
|---------------------------------------|---------------------|
| TensorFlow Developer Certificate      | Google              |
| AWS Certified Developer – Associate   | Amazon Web Services |
| Meta Front-End Developer Professional | Meta / Coursera     |
| Machine Learning Specialization       | Stanford / Coursera |
| MongoDB Certified Developer           | MongoDB University  |

